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Computer Vision Course

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Motivation & Objectives

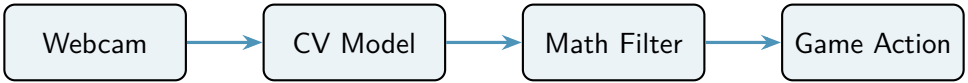
Goal: Build a fully client-side browser puzzle game controlled purely by hand gestures from a standard webcam.

Q1: The Classification Challenge

What is the most lightweight model sufficient for reliable hand gesture recognition? Does fusing Skeleton + Appearance help?

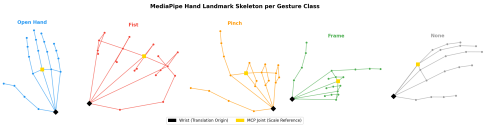
Q2: The Interaction Challenge

How do we filter natural hand jitter to bridge frame-level classifications into smooth, intentional game control?

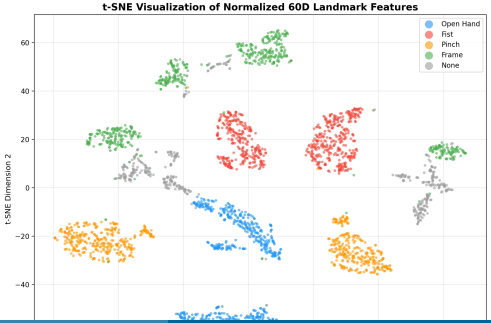


Classification Benchmark: Data & Features

Two Feature Streams



Skeleton (60D Vector via MediaPipe)



Dataset: HaGRID v2 (Subset)

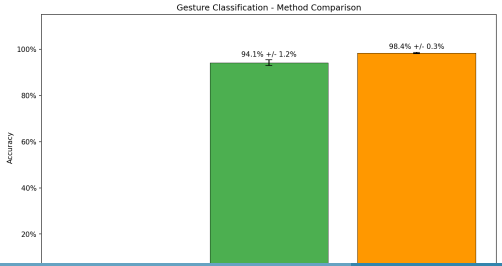
- 23,850 samples, 13,250 subjects
- 5 target classes (Pinch, Fist, Open, Frame, None)
- **Person-aware split**: No subject overlap between Train & Test

Evaluated Models:

- Heuristics (0 params)
- Random Forest (∼100K params)
- **MLP on Skeleton (60D, 14.7K)**
- CNN on Appearance (RGB, 1.5M)

Results & The "Ceiling Effect"

Method	Acc	Size
Random Forest	94.1%	8.6 MB
MLP (Skeleton)	**99.8%**	**227 KB**
CNN (Appearance)	100.0%	0.3 MB
Fusion	100.0%	0.5 MB



The Core Finding:

The 5 discrete classes are simply **too easy** for modern architectures.

- A tiny **14K-parameter MLP** essentially saturates the dataset (99.8%).
- **Fusion offered zero value**: Geometric features alone perfectly resolve spatial ambiguity.
- **Winner:** MLP on MediaPipe landmarks provides the ultimate lightweight engine.

The Real Challenge: Classification $\neq$ Control

We solved classification (99.8% accuracy).
But feeding raw frames directly to the game creates a terrible UX.

Hand Jitter
Natural micro-movements cause erratic cursor jumps.

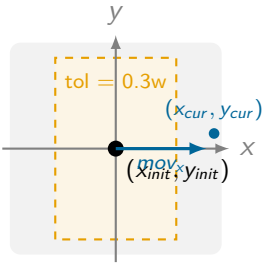
Transient Errors
A single misclassified frame drops the puzzle piece.

Lack of Intent
Users cannot hold perfectly still in 3D space.

"In order to make [the system] more suitable for disabled people... it needs to be more user-flexible... requires less energy" – Huda et al. 2025

Grid-Based Swipe Math (Huda et al. 2025)

Adapted the discrete directional **mathematical model** from wheelchair control to filter spatial noise and enable robust, low-energy puzzle swiping.



Logical Conditions:
Let $\Delta x = x_{cur} - x_{init}$, $\Delta y = y_{cur} - y_{init}$

Swipe Right (swap →):
$\Delta x > \mathbf{mov_x}$ AND $|\Delta y| < \mathbf{tol}$

Swipe Up (swap ↑):
$\Delta y < -\mathbf{mov_y}$ AND $|\Delta x| < \mathbf{tol}$

**Origin updates dynamically after swap for continuous multi-step swiping.*

Conclusion

Answers to Research Questions:

- **Q1 (Simplest Model):** The **227 KB MLP** on geometric skeleton data is the ultimate lightweight winner.
- **Q2 (HCI Bridge):** Explicit mathematical modeling (thresholds, tolerance zones) is essential to bridge ML probabilities into **reliable human-computer interaction**.

The Deployed System

MLP (227 KB)

$+$

Grid-Swipe tracker

Runs entirely client-side at 30+ FPS in any browser.

Thank You

Questions & Game Demo

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